# Conjecture 00000002783: a different equality pair for Banach–Mazur distance

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23 October 2026

## 1 The explicit necessary classification is false

Use the standard multiplicative Banach–Mazur distance between real finite-dimensional normed spaces,

$$d_{\mathrm{BM}}(X,Y) = \inf_{T:X\simeq Y} \|T\|\,\|T^{-1}\|,$$

where the infimum is over invertible linear maps with their actual operator norms. This convention is stated, for example, in Youssef, *Restricted invertibility and the Banach–Mazur distance to the cube*, printed p.7, before Corollary 3.3. The source's description using isometric embeddings is imprecise; we use this standard multiplicative distance, matching its proposed $\sqrt{n}$ bound, rather than a logarithmic or embedding distance.

In dimension two, take

$$E = \ell_2^2(\mathbf{R}), \qquad S = \ell_\infty^2(\mathbf{R}).$$

We prove $d_{\mathrm{BM}}(E,S) = \sqrt{2}$, but $E$ is linearly isometric to neither $\ell_1^2$ nor $\ell_\infty^2$. Thus this equality pair is not the named $\ell_1^2$–$\ell_\infty^2$ pair in either order. It refutes the explicit necessary equality classification. There is no need to assign a meaning to the additional second-extremal-value clause, or to prove or disprove the separate proposed all-pairs bound.

## 2 A lower bound for every invertible map

Fix any $T : E \to S$ invertible. Put

$$m = \|T\|, \quad M = \|T^{-1}\|, \quad u = T^{-1}(1,0), \quad v = T^{-1}(0,1).$$

The operator norm inequalities give $1 \le m\|u\|$ and $1 \le m\|v\|$. Both $(1,1)$ and $(1,-1)$ have sup norm one, so $\|u+v\| \le M$ and $\|u-v\| \le M$. The Euclidean parallelogram identity yields

$$2(\|u\|^2 + \|v\|^2) = \|u+v\|^2 + \|u-v\|^2 \le 2M^2.$$

Consequently

$$2 \le m^2(\|u\|^2 + \|v\|^2) \le m^2M^2, \qquad \sqrt{2} \le mM.$$

This is a bound over all invertible maps, not just the coordinate map.

## 3 Attainment and the genuine infimum

The coordinate identity $I : E \to S$ satisfies $\|I\| \le 1$ because $\max(|x_0|, |x_1|) \le \sqrt{x_0^2 + x_1^2}$. Its inverse satisfies

$$\|I^{-1}y\|_2^2 = y_0^2 + y_1^2 \le 2\|y\|_\infty^2,$$

so $\|I^{-1}\| \le \sqrt{2}$. The preceding universal lower bound forces $\|I\|\|I^{-1}\| = \sqrt{2}$. Therefore the actual set of norm products has $\sqrt{2}$ as its least element, and its infimum is $\sqrt{2}$. This also proves that the set is nonempty and bounded below; no default value of an empty or unbounded infimum is used.

## 4 Excluding both named isometry classes

Every Euclidean space satisfies the parallelogram identity, which is preserved by a linear isometry. In $S$, the coordinate vectors $a = (1, 0)$ and $b = (0, 1)$ have norms one, while both $a+b$ and $a-b$ have norm one. The parallelogram identity would read $2 = 4$. In $\ell_1^2$ the same coordinate vectors have norms one, while $a + b$ and $a - b$ have norm two, giving $8 = 4$. Thus $E$ admits a linear isometry equivalence to neither space. This rules out the named unordered pair in either direction, not merely an equality of implementation types.

## 5 Formal objects and verification

Lean uses Mathlib's actual `EuclideanSpace` over `Fin 2`, its sup-norm function space `Fin 2 → R`, and the actual `PiLp 1` space. The two example spaces have proved real dimension two. `distortion` uses operator norms of each actual `ContinuousLinearEquiv` and its inverse. `distances` is the range over all those equivalences, and `bmDistance` is its `sInf`. `distortion_isLeast` proves both membership and the universal lower bound before `bmDistance_eq` uses the infimum theorem. The coordinate map is an actual continuous linear equivalence.

The formal `NamedPair` explicitly includes both orderings of the two named linear-isometry classes. `no_unique_equality_classification` negates the asserted classification, and `counterexample` bundles the dimensions, actual distance and both non-isometry results.

Lean 4.33.0 and Mathlib commit `db584cd6d46c92f209a44c0f1c829460d327499d` are pinned. Source-bound receipts record a new Main build, transitive axiom audits and full official `leanchecker --fresh Main` replay. No `sorry`, `native_decide` or added axiom is used. OpenAI Codex assisted; a separate supervising Codex agent reviewed the formal chain. This is AI review, not human certification. Organizer acceptance is pending.